Vidush Singhal

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West Lafayette, IN, USA (Open to full time roles)

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Summary

Ph.D. student in Electrical and Computer Engineering at Purdue University specializing in compiler infrastructure, program analysis, and performance optimization. Experienced with LLVM/Clang, MLIR/Torch-MLIR, GPU compiler tooling, automatic parallelization, data layout optimization, and software verification.

Experience

Microsoft Research (RiSE) — Redmond, WA
Research Intern — May 2025 – Aug 2025

- Built `C2Pulse`, an LLVM/Clang-based transpiler for verifying memory correctness of annotated C programs.
- Implemented a Clang-to-Clang source transformer (~**7K LOC of C++**) that emits Pulse code for F* verification.
- Built a C benchmark suite and verified C components of DPE (Dice Protection Environment) through C2Pulse.

Lawrence Livermore National Laboratory (LLNL) — Livermore, CA
Computing Scholar Intern — May 2024 – Aug 2024

- Engineered an LLVM Attributor pass to optimize byte layout within stack allocations for improved spatial locality.
- Implemented LLVM IR analyses and transformations to eliminate dead bytes and reorder frequently accessed bytes.
- Reduced XSBench runtime by **6%** when running under the sanitizer.

Nod.ai Labs (Acquired by AMD) — Santa Clara, CA
Machine Learning Compiler Engineering Intern — May 2022 – Aug 2022

- Implemented Torch-MLIR kernels and compiler support for multiple PyTorch operators.
- Improved the PyTorch-to-MLIR compilation flow by expanding operator coverage.
- Added support for the Facebook DLRM recommendation model in the PyTorch-to-MLIR pipeline.
- Investigated ILP solver improvements for the Alpa scheduling framework.

Purdue University - PLCL Group — West Lafayette, IN
Graduate Research Assistant — May 2021 – Present

- **Gibbon Compiler**
  - Optimized layouts of recursive datatypes to improve spatial locality.
  - Measured **speedups of 1.14x to 54x** over the best prior work.
  - Implemented an array-of-structs to structure-of-arrays-style transformation for packed algebraic datatypes.
  - Enabled selective traversal with **1.1x to 12x speedups** and ongoing loopification/vectorization optimizations.
- **Orchard Compiler**
  - Developed a compiler framework to automatically fuse and parallelize tree traversals.
  - Measured **1x to 5x speedups** over baseline tree traversals.
- **Cornucopia**
  - Developed a fuzzing-based framework to generate optimized binary corpora using `llc` optimization flags.
  - **Discovered bugs** in LLVM compiler optimization passes.
  - **Uncovered decompilation failures** in angr, Ghidra, and related binary analysis tools.
  - Embedded ground-truth metadata in generated binaries to evaluate binary analysis tools.
- **Copse Compiler**
  - Developed a framework for vectorized evaluation of fully homomorphic encryption (FHE) decision trees.

Senior Design Project - SoCET Team — West Lafayette, IN
Undergraduate Research — Jul 2020 – May 2021

- Developed an LLVM-based compiler tool for proprietary RISC-V hardware.
- Implemented multiple LLVM RISC-V backend Machine IR passes to identify sparse program regions.
- Annotated sparse regions so hardware could detect them at runtime and perform runtime checks.

Selected Projects

- **Global Tensor Layout Optimization for IREE** (June 2026): Explored global tensor layout optimization in the IREE compiler pipeline.
- **LLM-Assisted Specification Inference for Verus** (Fall 2025): Explored LLM-assisted inference of Verus specifications using symbolic reasoning tools.
- **OpenMP GPU Offloading for Parallel STL Algorithms** (Fall 2024): Built a proof-of-concept evaluation of OpenMP GPU offloading for C++17 STL-style algorithms.

Education

Purdue University, West Lafayette, IN
Ph.D. in Electrical & Computer Engineering — Expected December 2026
M.S. in Electrical & Computer Engineering — May 2023
B.S. in Computer Engineering — May 2021
Relevant Coursework: Compiler Code Generation and Optimization, Compilers for GPUs, Programming Parallel Machines, Reasoning About Programs, Programming Languages, Formal Languages/Computability/Parallelization, Holistic Software Security

Publications

- Raghav Malik, **Vidush Singhal**, Benjamin Gottfried, Milind Kulkarni. "Vectorized Secure Evaluation of Decision Forests." *PLDI 2021.*
- **Vidush Singhal**, Akul Abhilash Pillai, Charitha Saumya, Milind Kulkarni, Aravind Machiry. "Cornucopia: A Framework for Feedback Guided Generation of Binaries." *ASE 2022.*
- **Vidush Singhal**, Chaitanya Koparkar, Joseph Zullo, Artem Pelenitsyn, Michael Vollmer, Mike Rainey, Milind Kulkarni. "Optimizing Layout of Recursive Datatypes with Marmoset." *ECOOP 2024.*
- **Vidush Singhal**, Laith Sakka, Kirshanthan Sundararajah, Ryan Newton, Milind Kulkarni. "Orchard: Heterogeneous Parallelism and Fine-grained Fusion for Complex Tree Traversals." *ACM TACO 2024.*
- Chaitanya S. Koparkar, **Vidush Singhal**, Aditya Gupta, Mike Rainey, Michael Vollmer, Artem Pelenitsyn, Sam Tobin-Hochstadt, Milind Kulkarni, Ryan R. Newton. "Garbage Collection for Mostly Serialized Heaps." *ISMM 2024.*

Skills

Languages: C, C++, Haskell, Python, Java, Bash, CUDA, Assembly (ARM, x86), SystemVerilog
Compilers & Verification: LLVM, Clang, MLIR/Torch-MLIR, GCC, Coq, Dafny, F*, Pulse
Parallel, Systems & ML: OpenMP, MPI, Pthreads, OpenCilk, Intel TBB, gem5, AFL++, GDB, Linux, PyTorch, TensorFlow, NumPy

Awards, Grants & Service

Dean's List and Semester Honors, Purdue University (2017 - 2020)
LLVM Developers' Meeting Travel Grant 2024
Research Service - AEC/PC/Reviewer for PLDI, CGO, ICFP, PPoPP, POPL, SPLASH, C3PO, JOSS (2023–2026)
Seminar Co-Coordinator, Purdue Programming Languages and Systems Research Group (2025 - present)